Intoduction to GeoMechanics

Case Study

Well Name - Chandon-2

Group - F

Instructor

Prof. Partha Pratim Mandal

Overview of Well

Location

Chandon gas field is on the
Exmouth Plateau,
part of the North Carnarvon Basin,
northwest Australia.

Well, Borehole Operator

Chevron Australia Pty Ltd

Drilled By

Ensco Offshore

Depth

Chandon-2 was drilled vertically
in 1168 m
of water (below mean sea level)

Objectives

Perform basic wireline log data (relevant for geomechanical study – gamma ray, density, sonic, resistivity, porosity) quality control and replacement of outlier with moving average filter operation

Define lithology based on Vshale cut-off (Vshale <=0.4) into shale and sand

Estimate In-situ stress at depth

Predict Pore pressure using suitable method or formula (e.g. Eaton's, Bower's etc.)

Create SHmax orientation map from world stress map project.

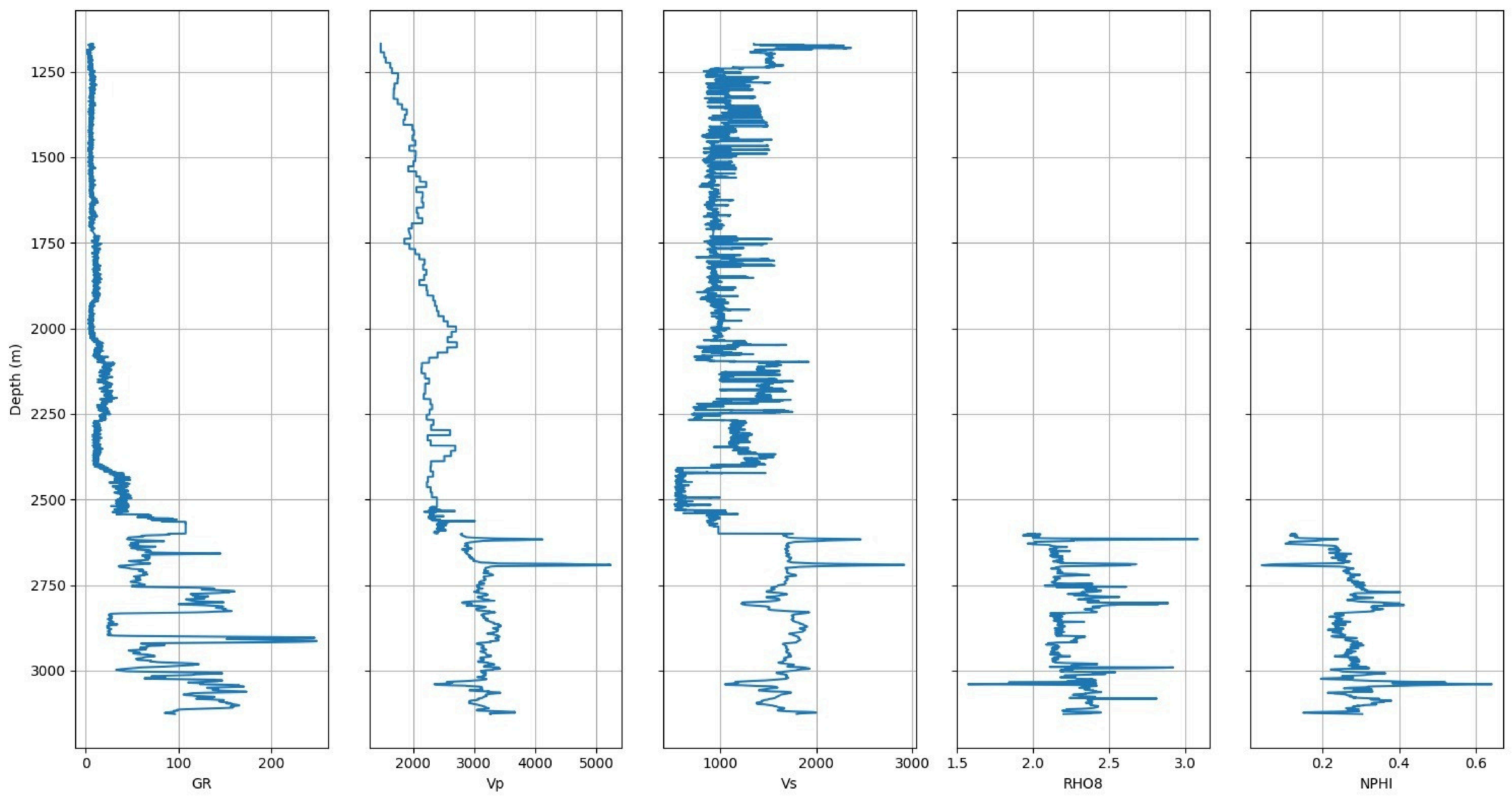
Estimate different Rock Mechanical Properties (Young's Modulus, Poisson's ratio, Shear Modulus, UCS, etc.)

Calibrate the Dynamic rock mechanical properties with Static one, establish a relation between static and dynamic properties.

Build 1D geomechanical model and provide statistical distribution of the mechanical properties of the reservoir.

Outline of QC

Given Raw Data

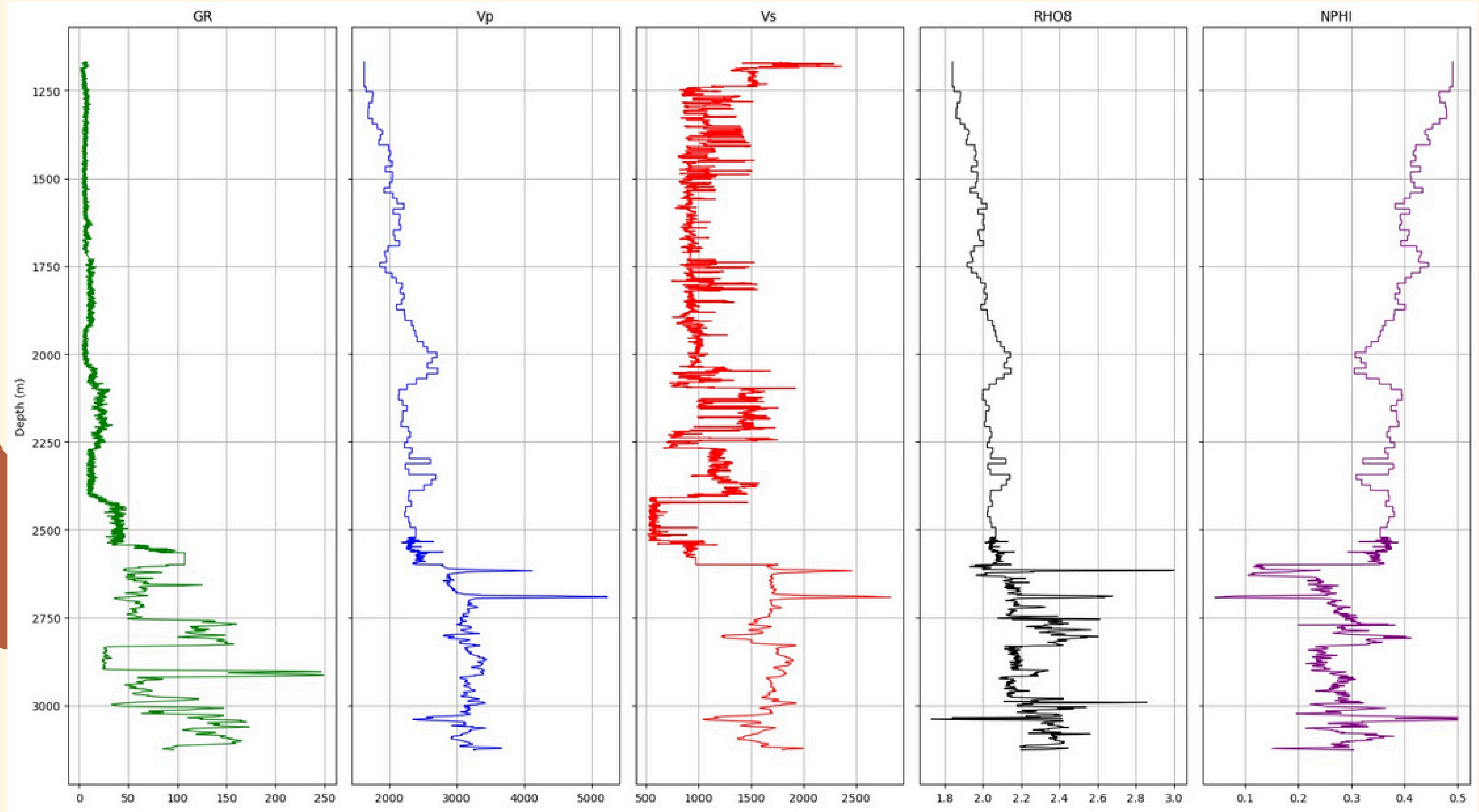


We used Gardner's Equation for Estimating the missing RHO8.

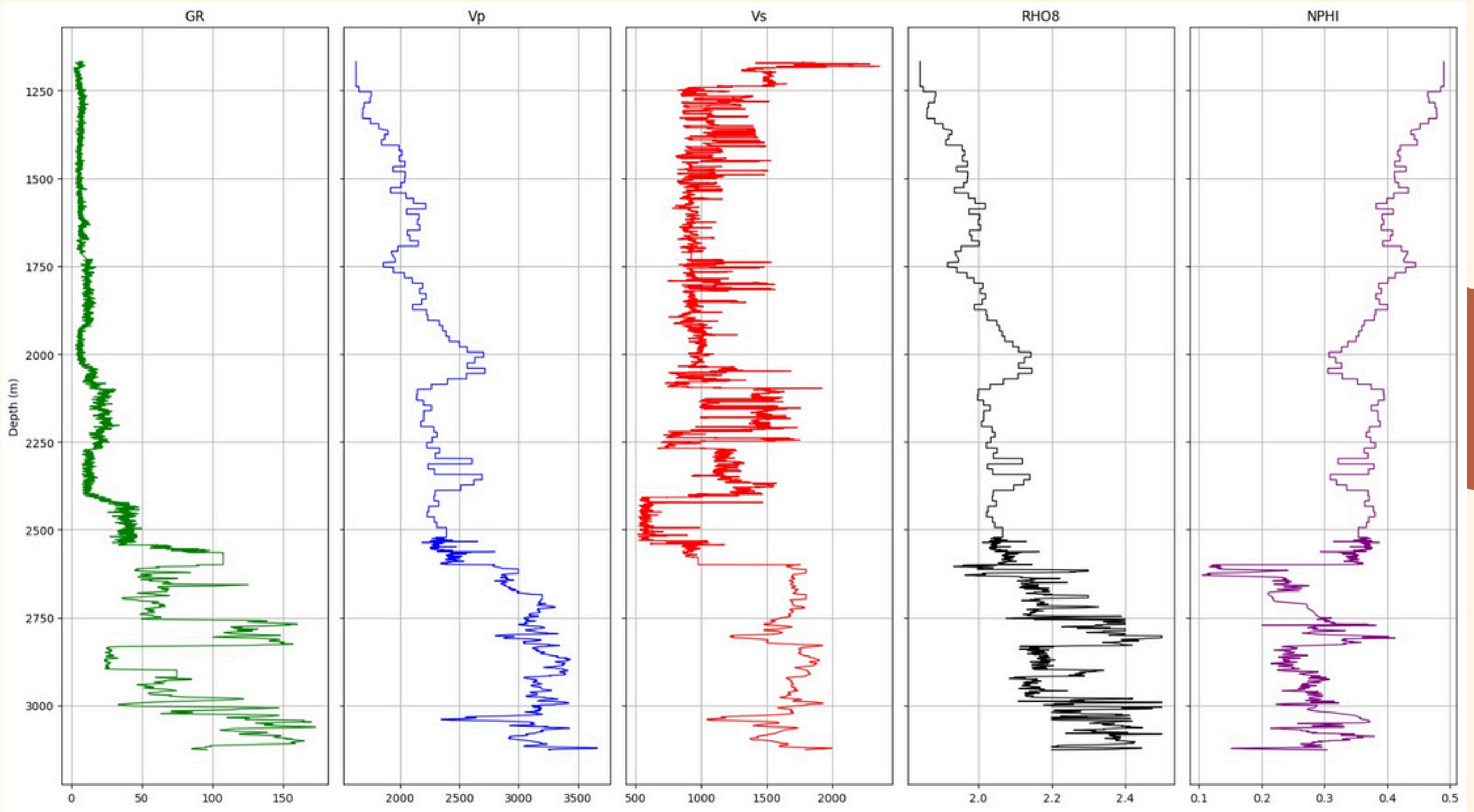
$$\rho = \alpha V_p^{\beta}$$

alpha=1.2195
beta=0.0780

Spikes Removal



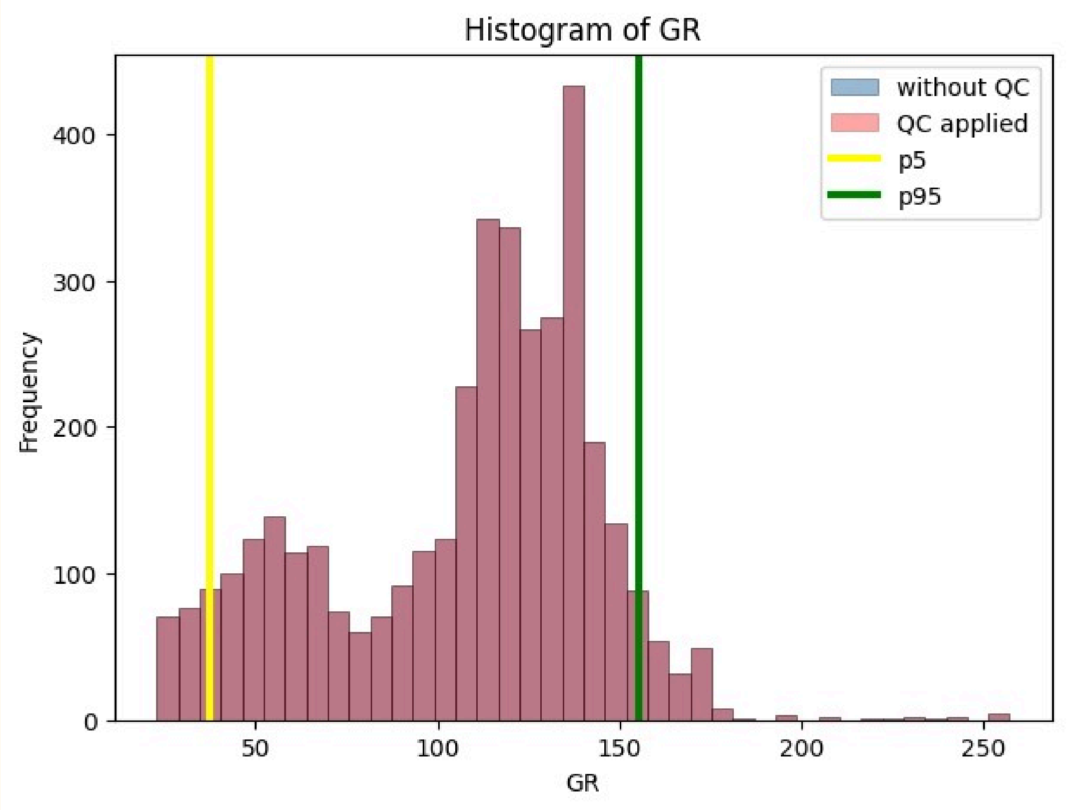
Data with spikes

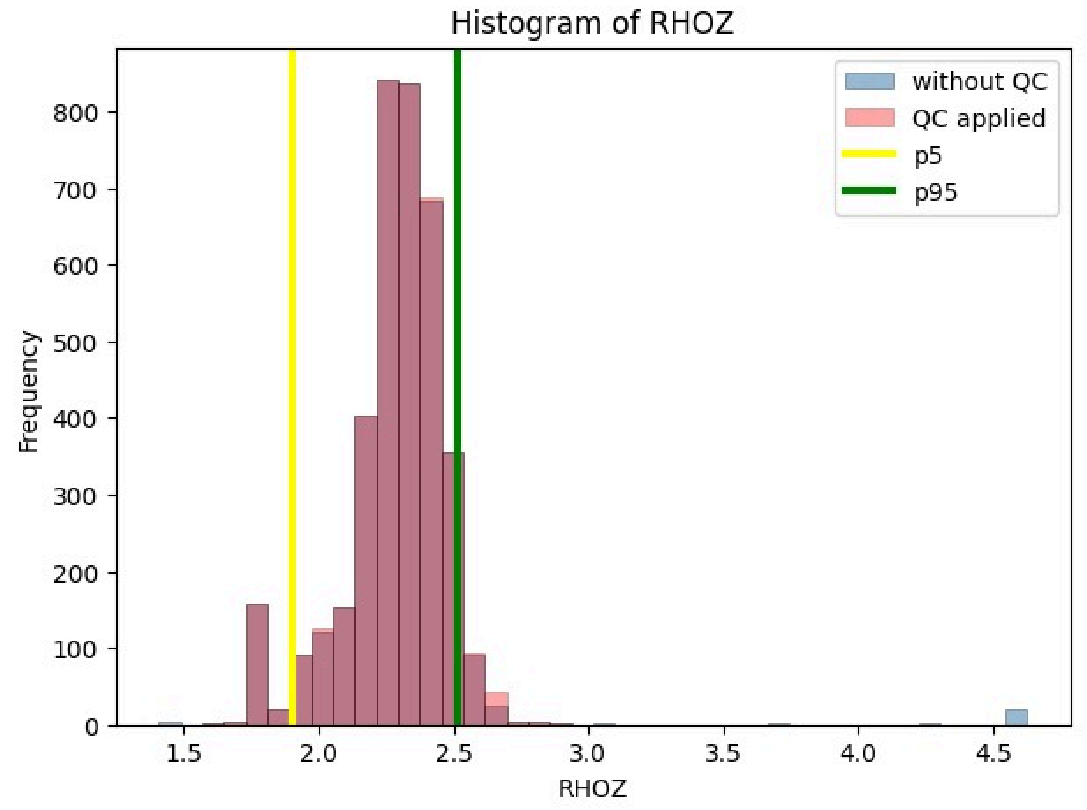


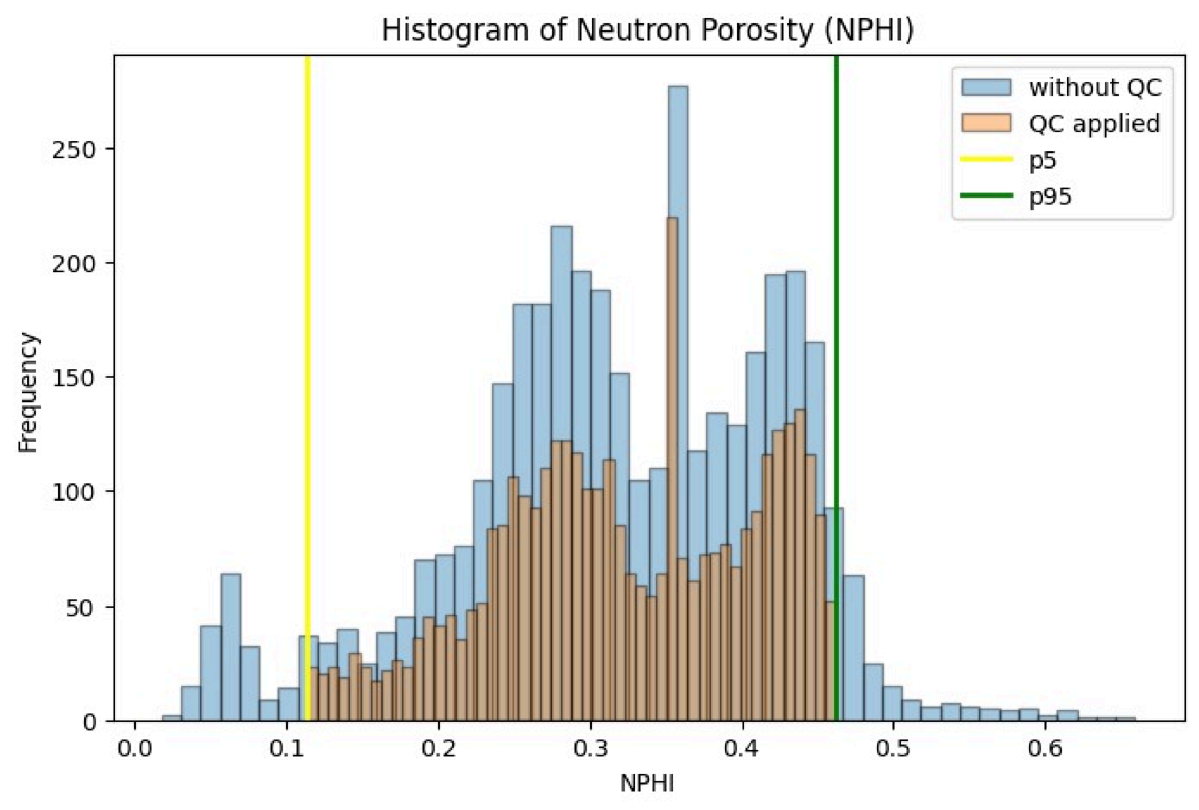
Data without spikes

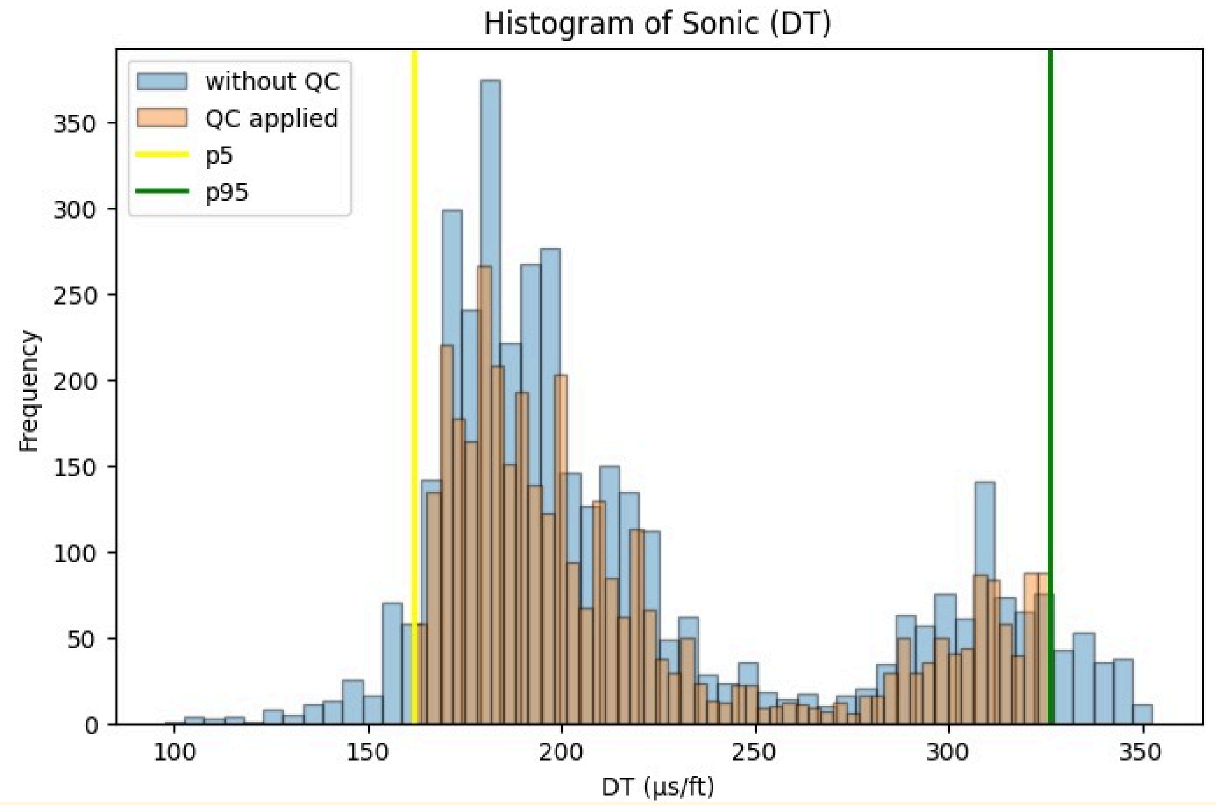
We get Missing density
from which we calculated bulk density Porosity

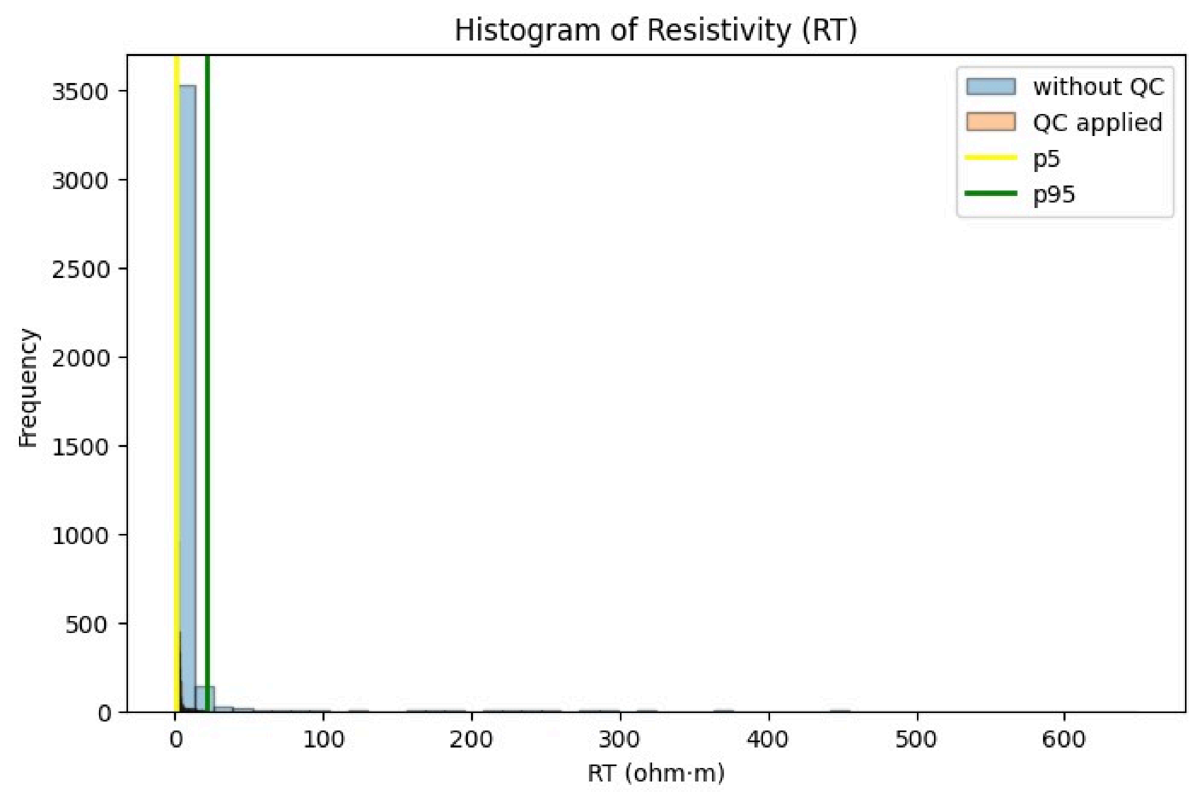
Formation-wise QC



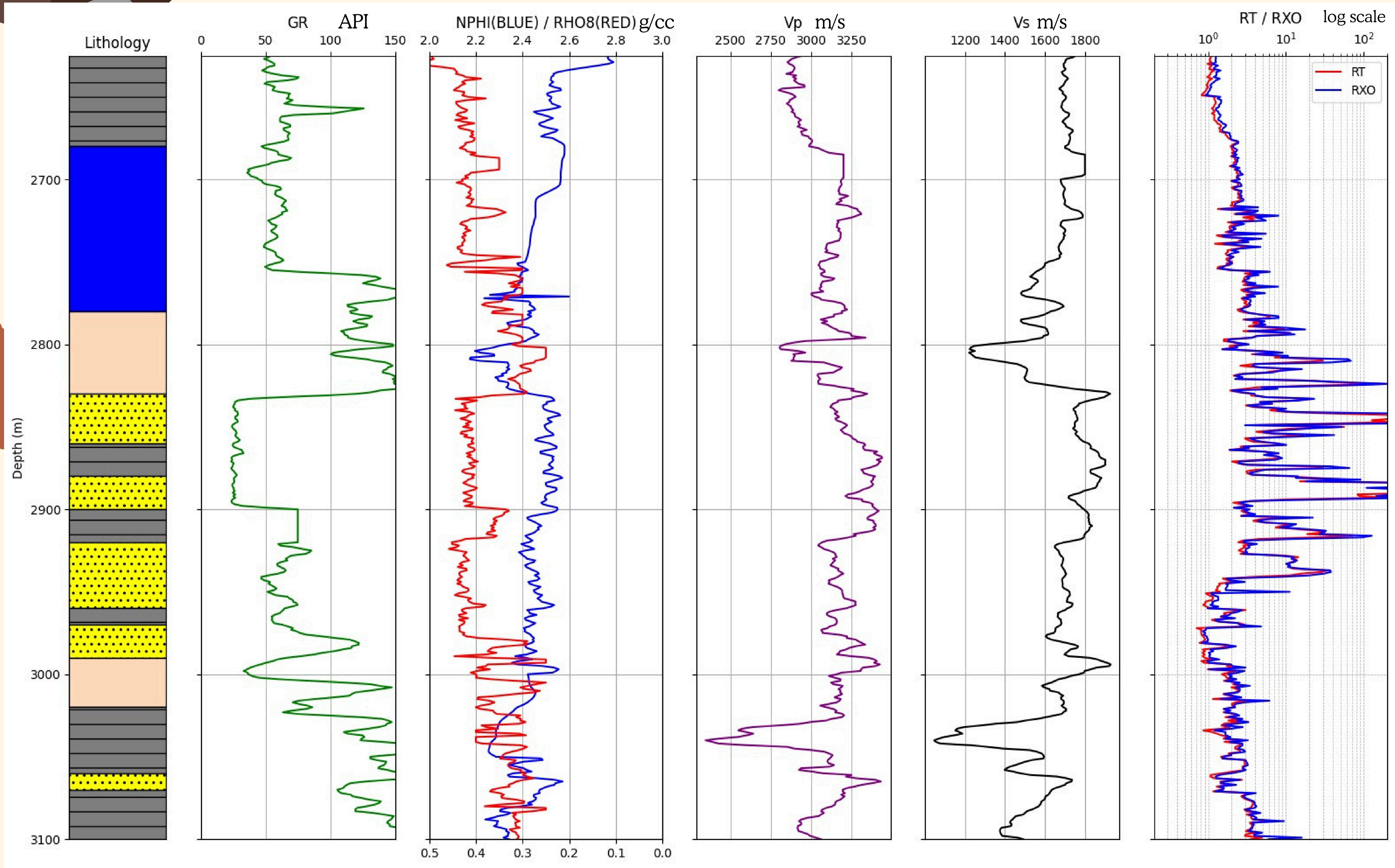








Lithology



Lithology Legends

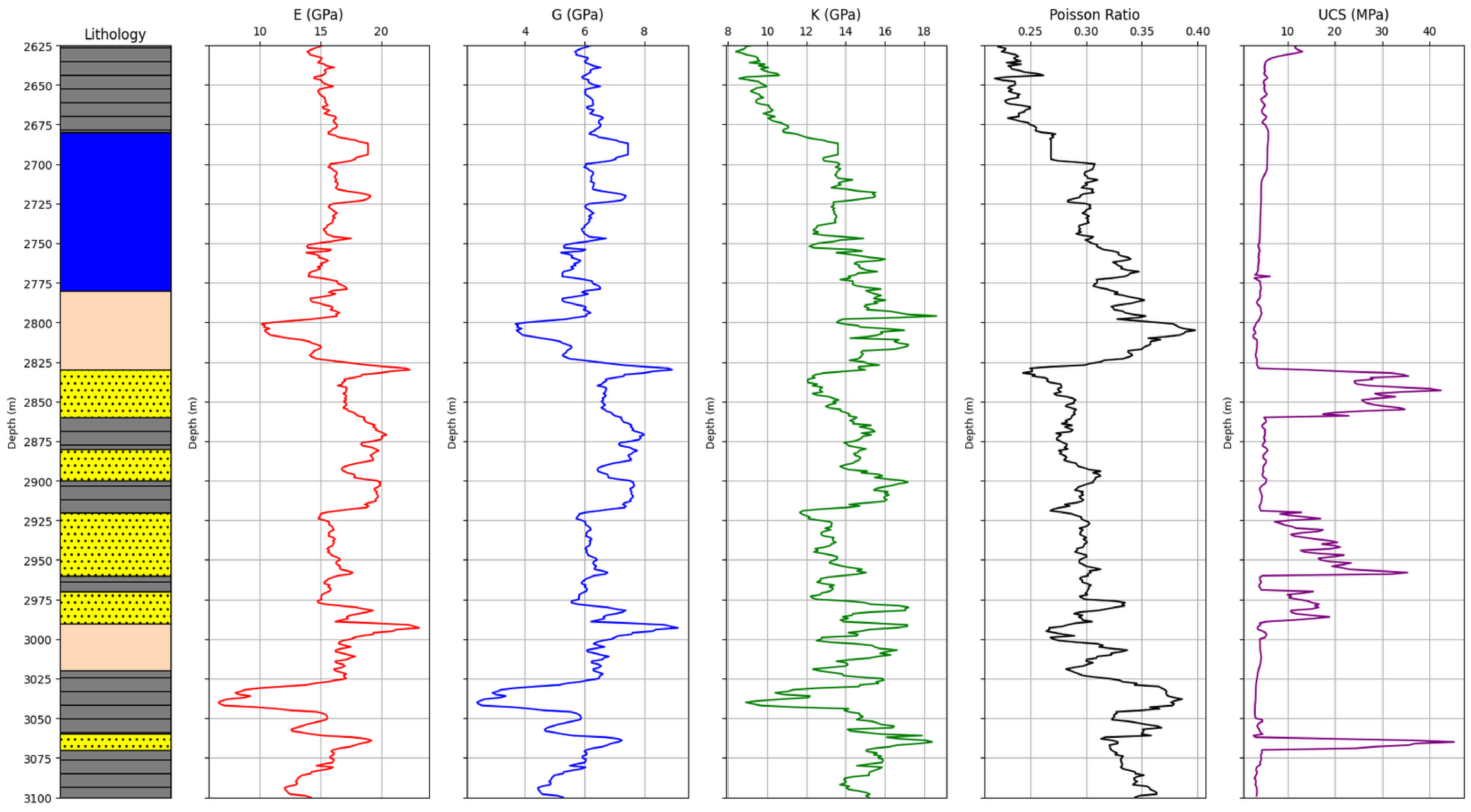
Blue-Marl

Grey-Shale

Bluff pink-silt stone

Yellow-Sandstone

Elastic Modulus



UCS

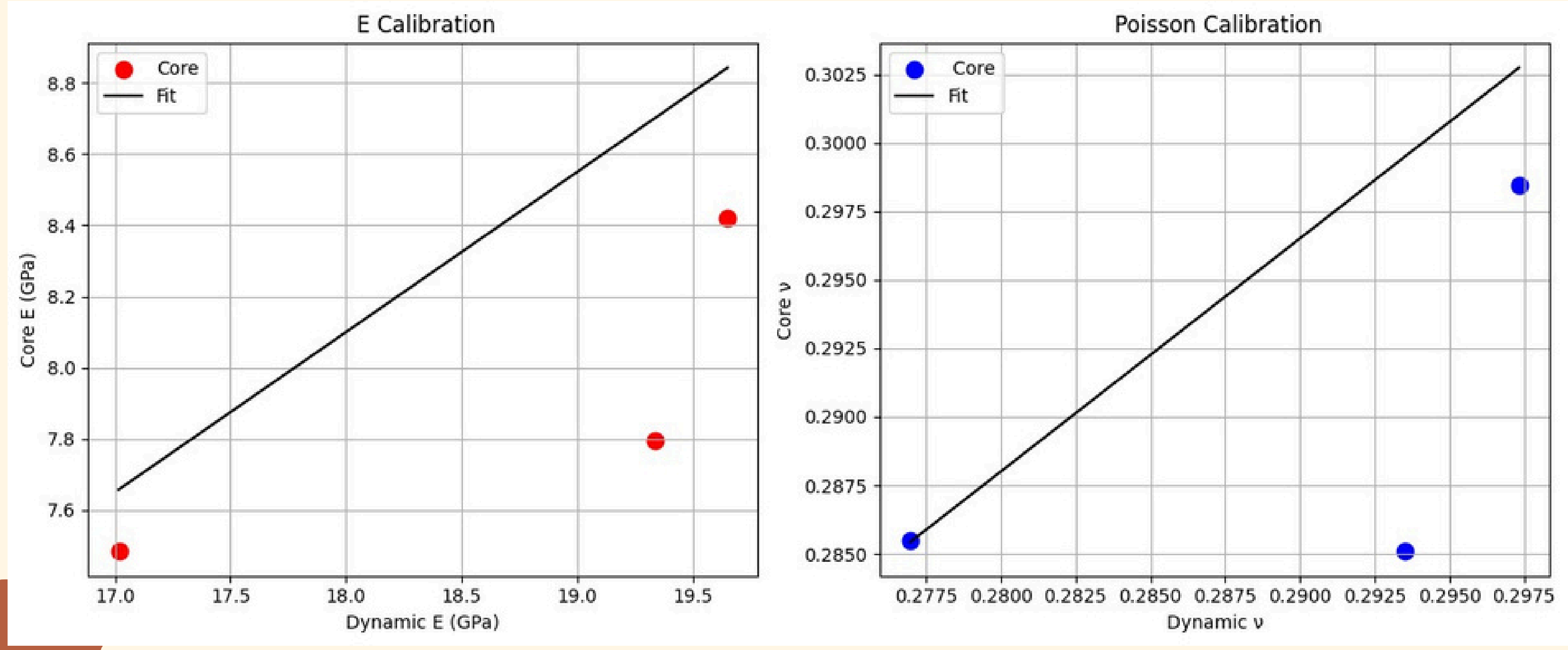
FOR SAND

$$UCS = 254\,(1-2.7\phi)^2$$

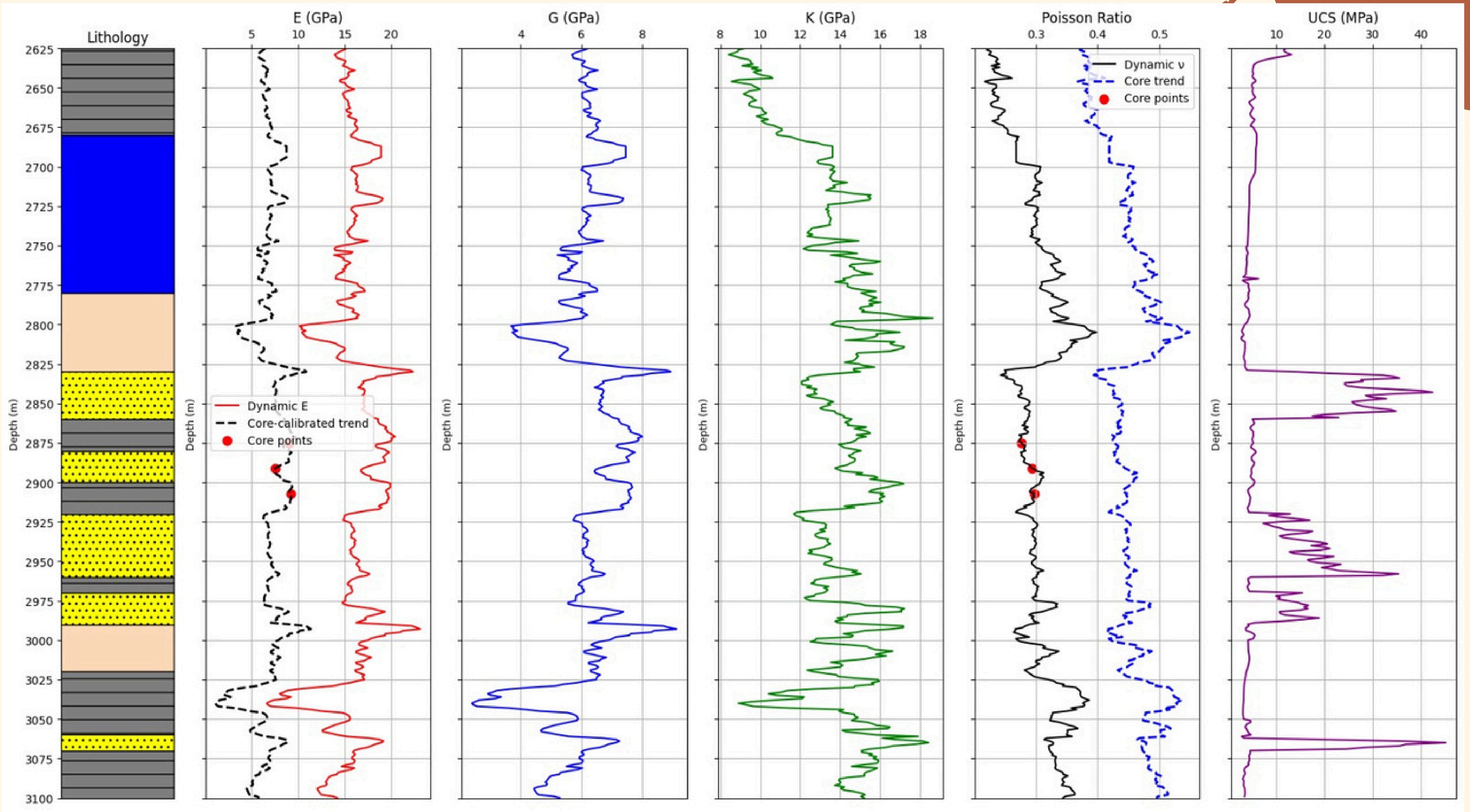
FOR SHALE

$$UCS = 1.001 \cdot \phi^{-1.143}$$

Dynamic vs Static



Using simple Regression model,
we fitted our Static and Dynamic data

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Final E equation:
E_core = 0.620 * E_dynamic + -2.978
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Final nu equation:
nu_core=1.00*nu_dynamic+0.30
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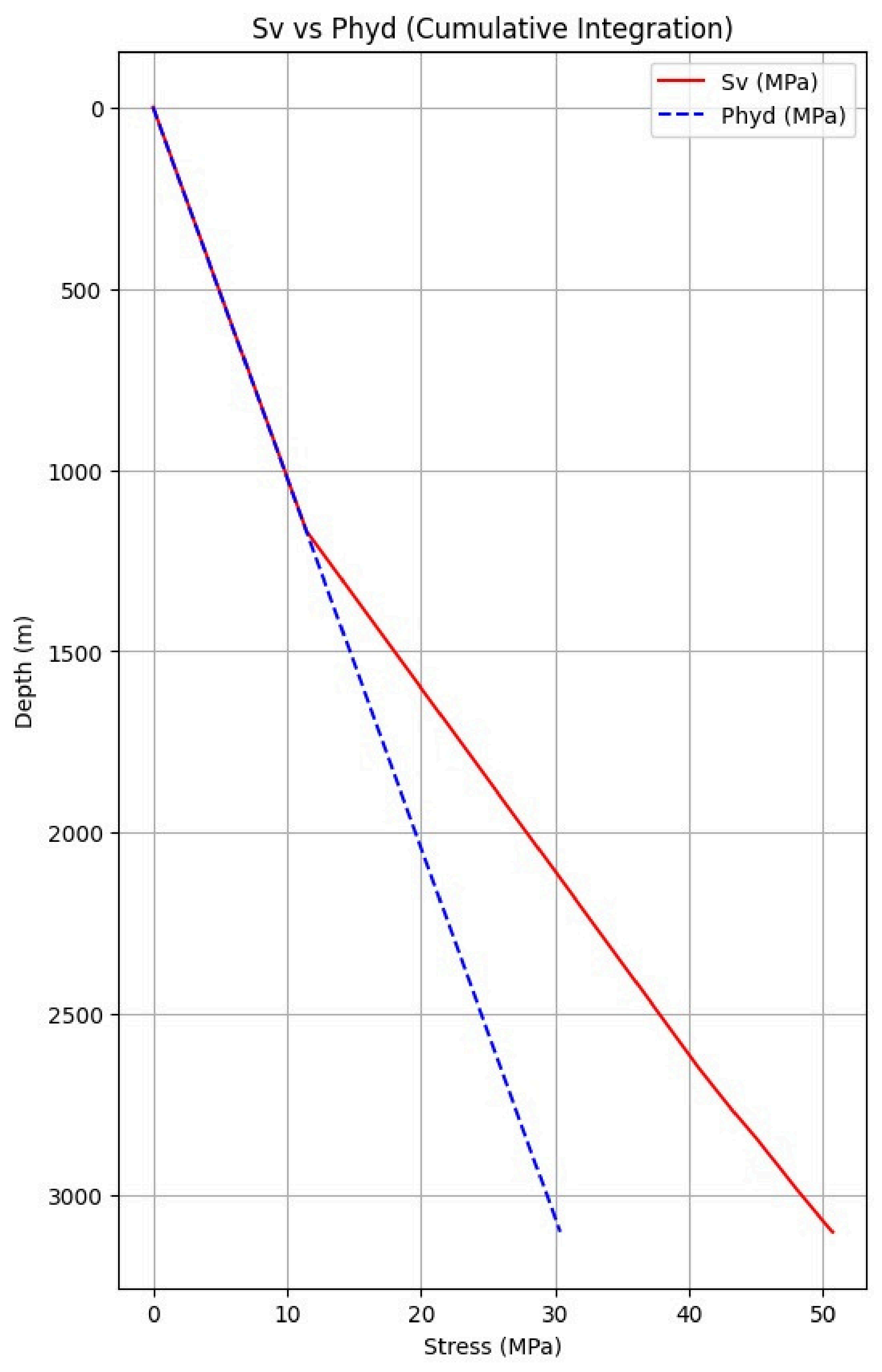
Calculation of Sv & P-hyd



Our sea bed was at 1168m

from 0 to 1168m

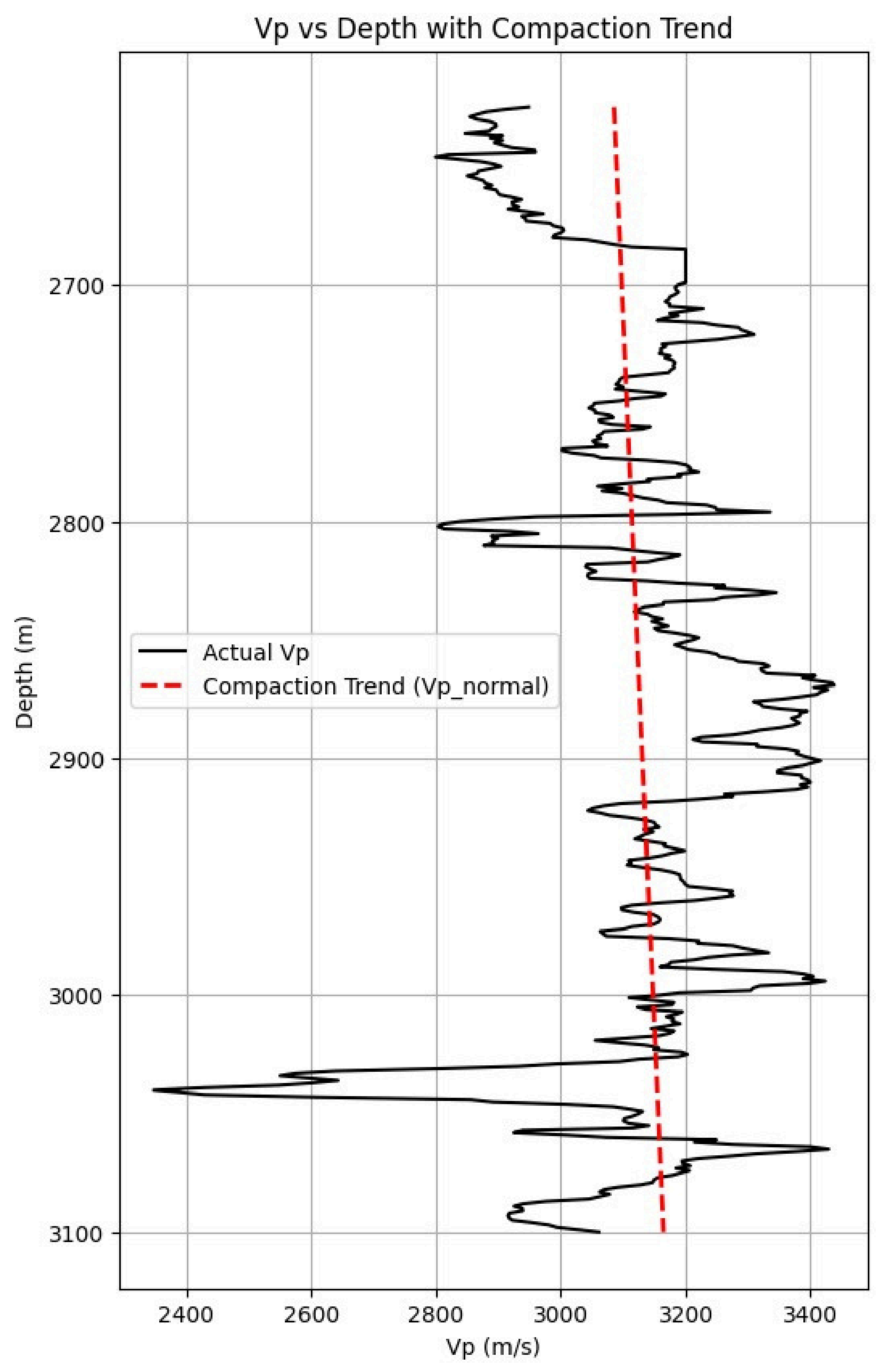
PHYD=SV

from 1168 to 3100m

Phyd=1000*9.8*h

Sv=Phyd+ RHO8*9.8*H

Normal Compaction Trend (NCT)



For Applying the eaton's formulae

we need vp(normal)

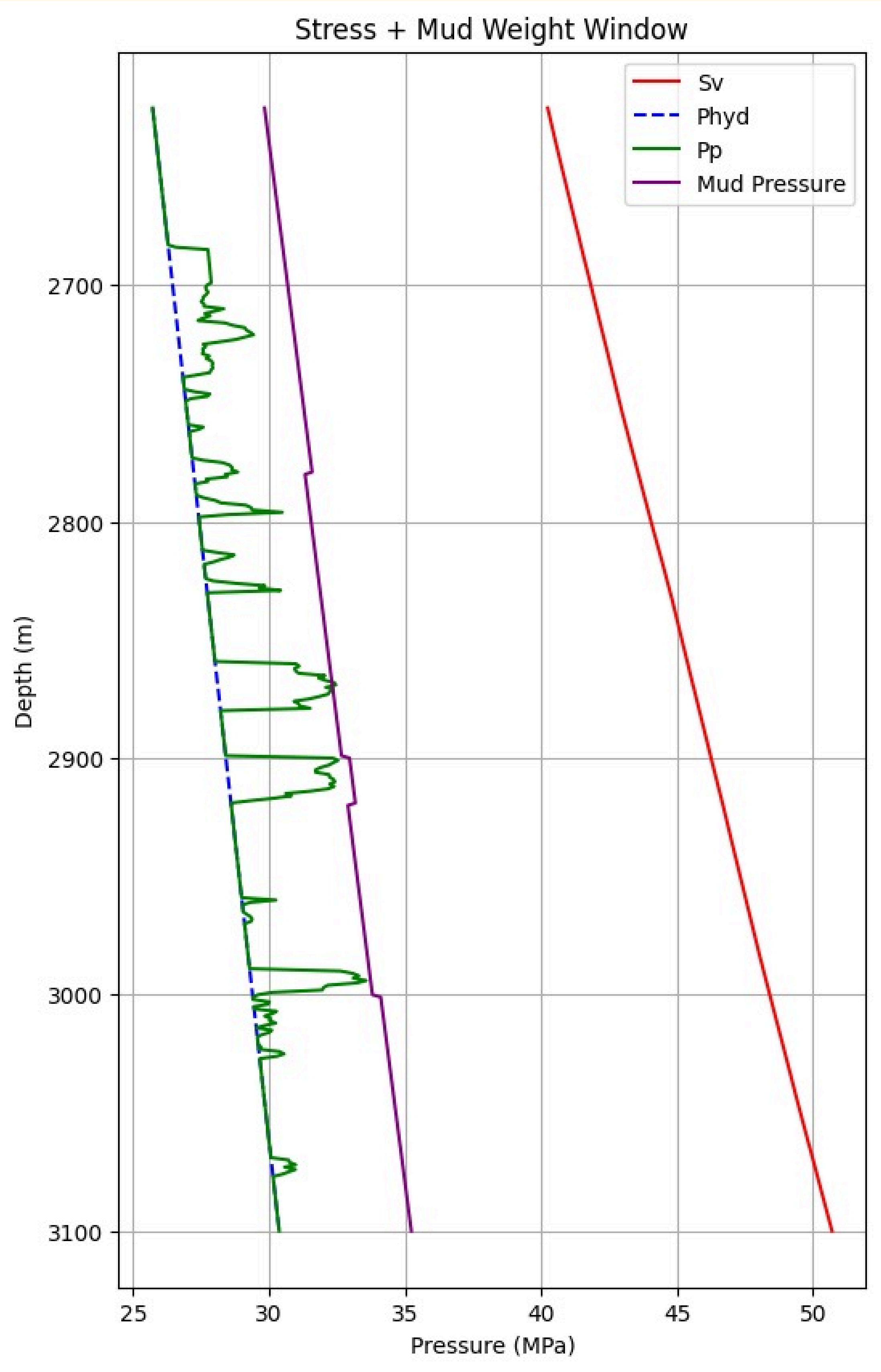
Vp(Normal)= $a \cdot e^{bx}$

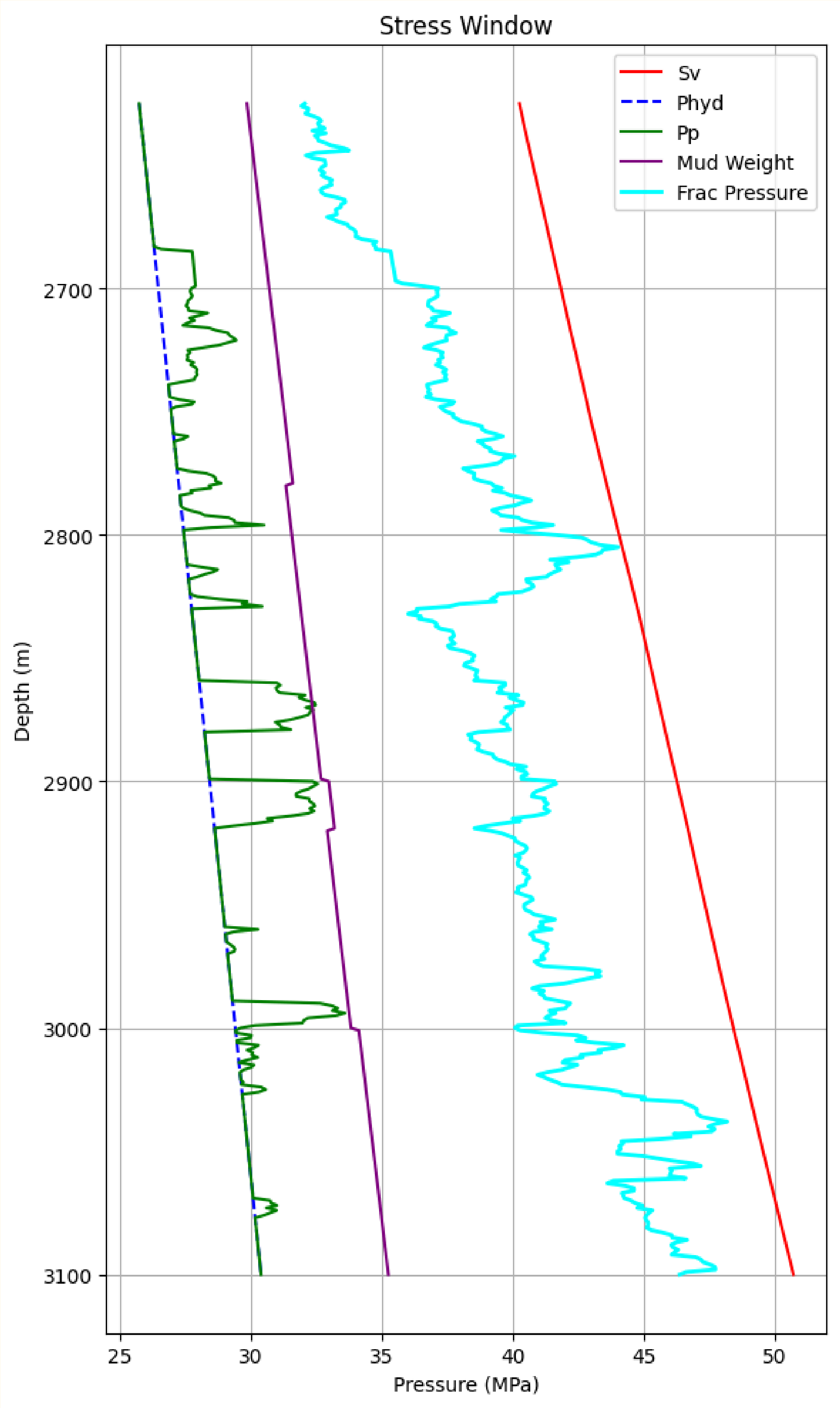
after fitting we got

a = 3085.15

b = 0.025

Estimation of Pore-Pressure





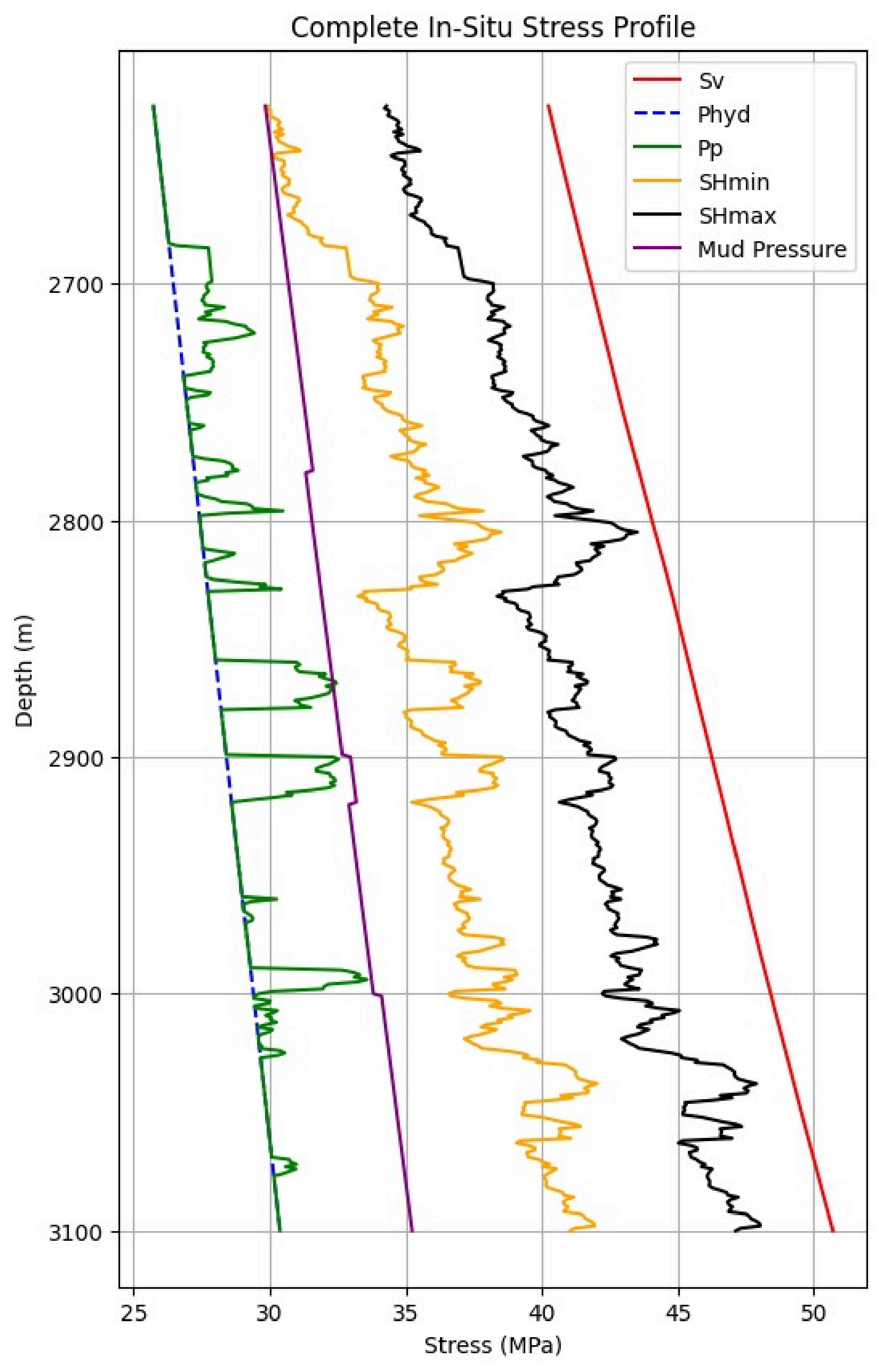
$$P_p = S - (S - P_h) \times \left(\frac{V_{p_o}}{V_{p_n}} \right)^E$$

We took E =3.0 which showed good variation in pore pressure

S=Sv

Ph=Phyd

Estimation of SH-min & SH-max



We calculated Shmin

$$S_{hmin} = k(\sigma_v - P_p) + P_p$$

K=V/(1-V)

V=POISSON RATIO

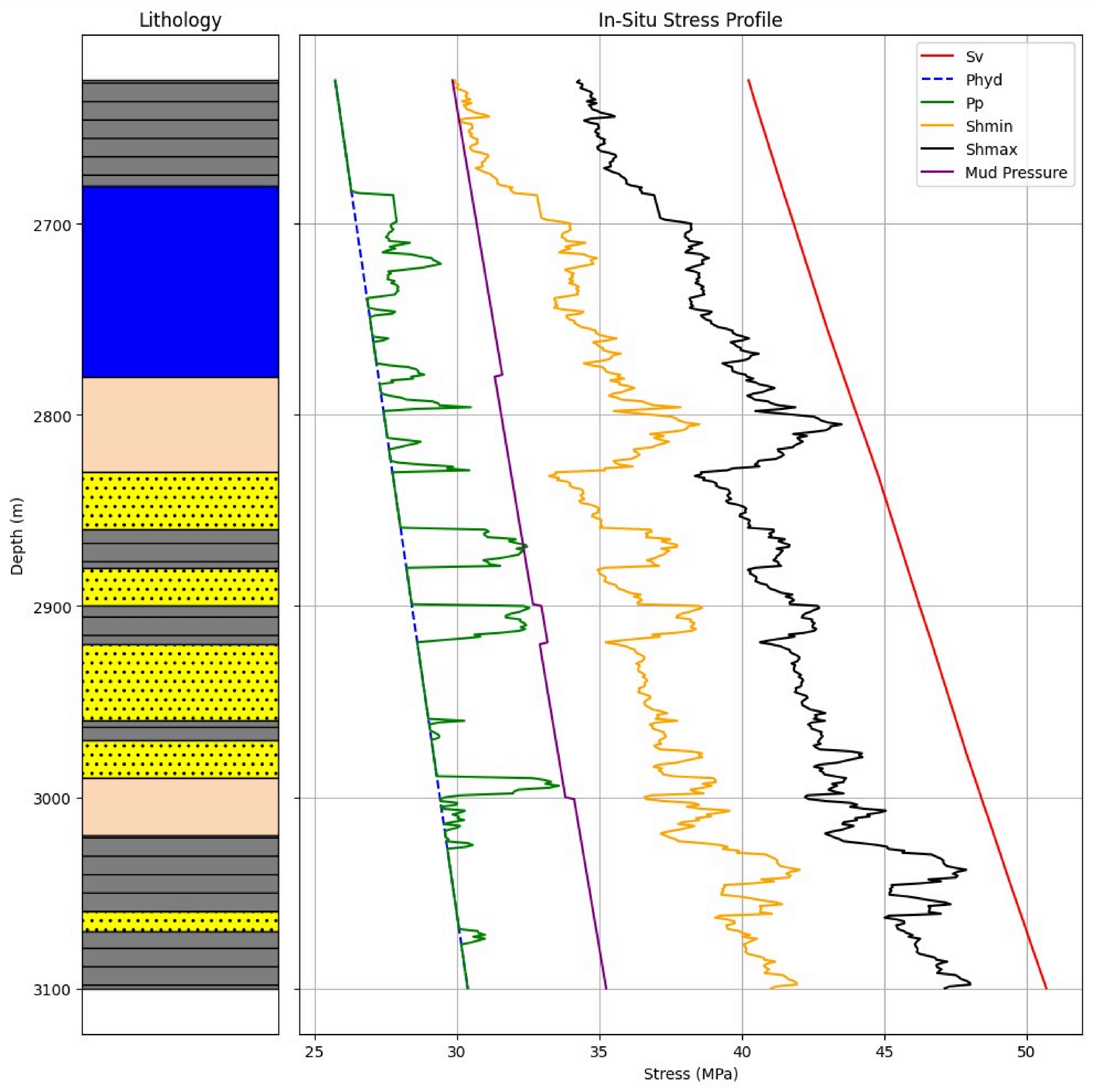
SHMAX

$$S_{Hmax} = (K_0 + \alpha)(S_v - P_p) + P_p$$

ALPHA=3

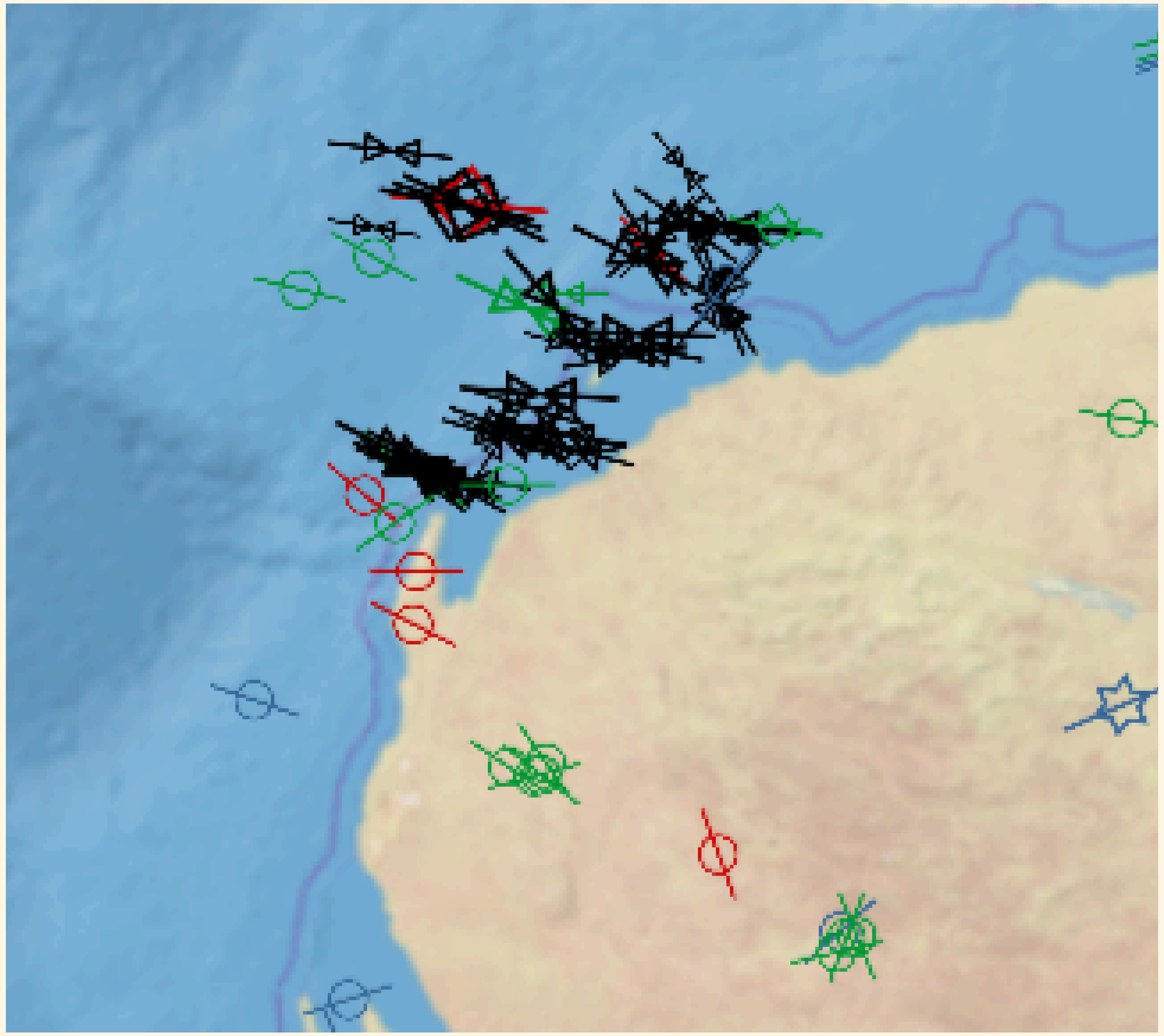
IT IS TECTONIC FACTOR

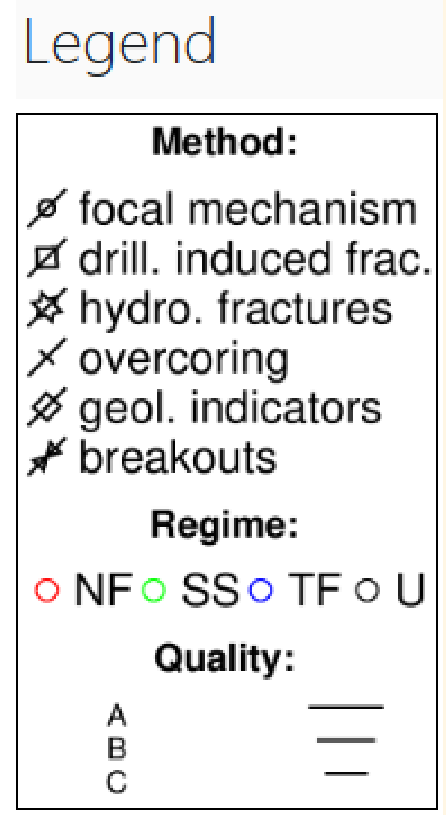
IN-SITU STRESS WITH LITHOLOGY

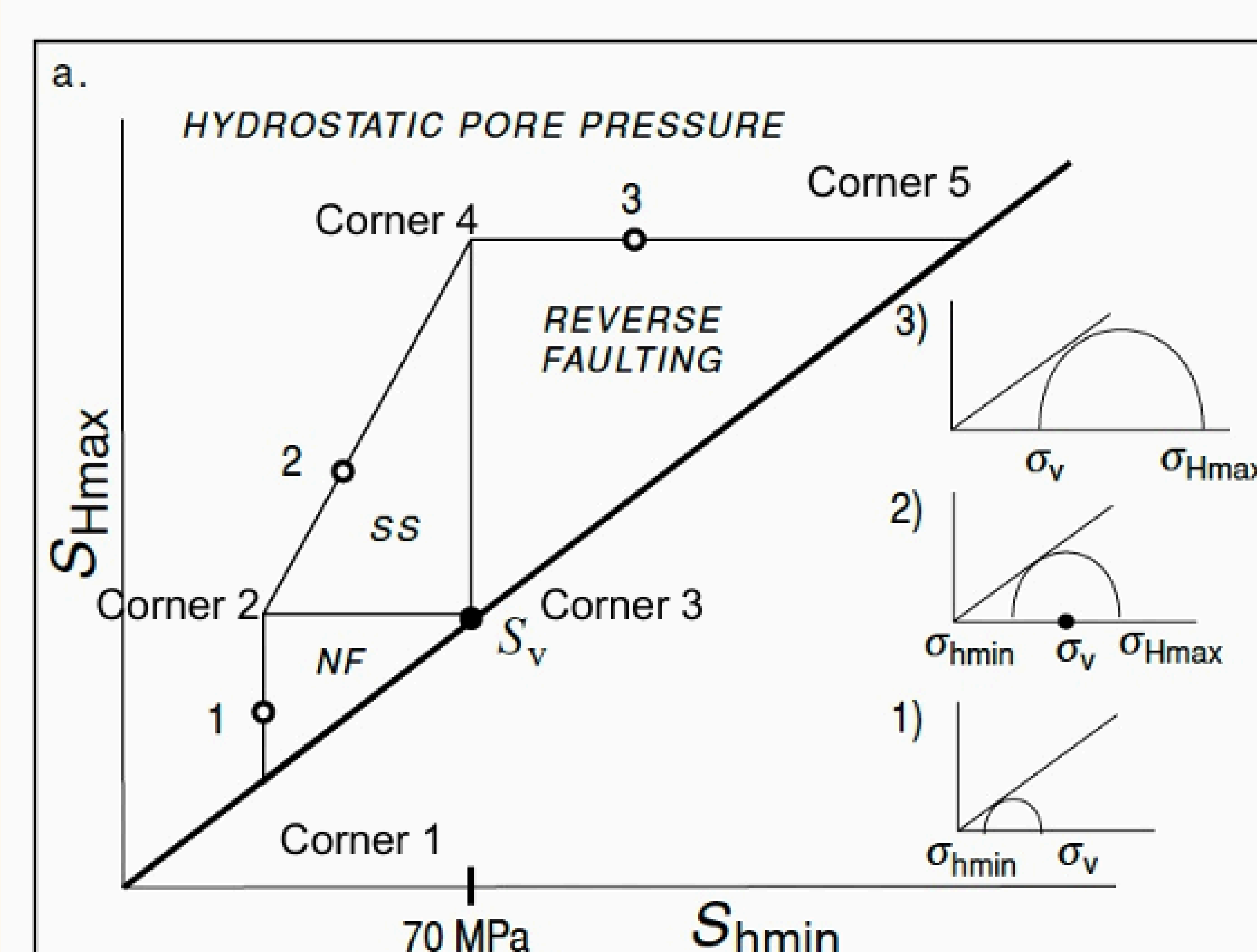


WE OBSERVE THE PORE PRESSURE IN SHALE REGIONS

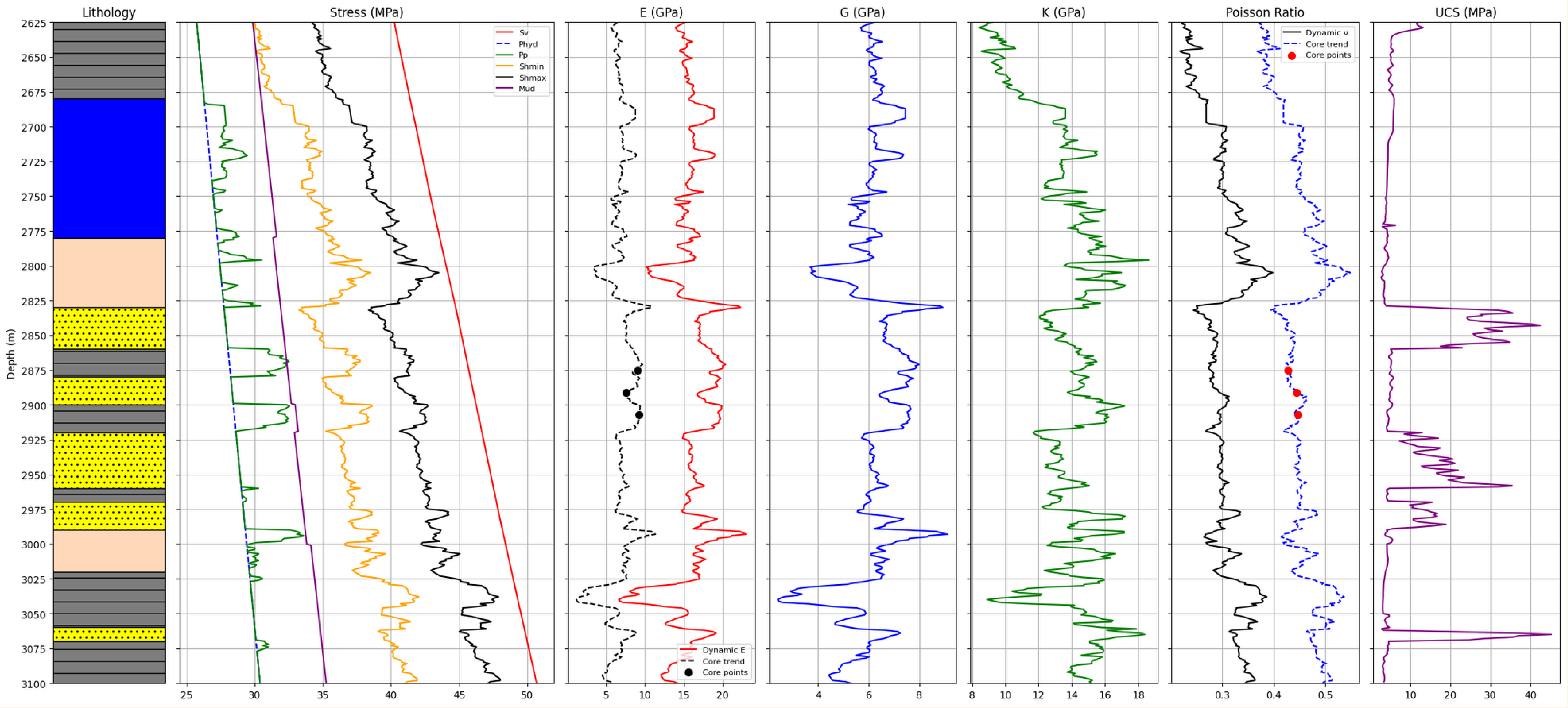
SH-MAX ORIENTATION







1-D Geomechanical Model



TEAM MEMBERS:

1. NIKHIL KUMAR

3.ABHI SAKARIYA

2.NABA FATIMA

Thank You